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Comparative Omics-Based Identification and Expression Analysis of a Two-Component System in *Vigna radiata* in Drought Stress

Muhammad Amin Afzal ¹, Farrukh Azeem ¹ , Shumaila Afzal ², Naila Afzal ¹, Muhammad Rizwan ³ , Hyojin Seo ⁴, Asad Ali Shah ^{1,*} and Muhammad Amjad Nawaz ^{5,*} 

¹ Department of Bioinformatics and Biotechnology, Government College University, Faisalabad 38000, Pakistan; azeuaf@hotmail.com (F.A.)

² Department of Microbiology, University of Agricultural, Faisalabad 38000, Pakistan

³ Department of Environmental Sciences, Government College University, Faisalabad 38000, Pakistan

⁴ Korean Soybean Research Institute, 950 Worasan-ro, Jinju 52840, Republic of Korea

⁵ Advanced Engineering School (Agrobiotek), Tomsk State University, Lenin Ave, 36, 63050 Tomsk, Russia

* Correspondence: asadshah.12@gmail.com (A.A.S.); amjad_ucauos@yahoo.com (M.A.N.)

Abstract: Two-component system (TCS) genes regulate a wide range of biological activities in prokaryotes and eukaryotes, including plants. TCS plays an important role in cellular responses to external stimuli, such as biotic and abiotic factors. In plants, this system supports cell division, leaf senescence, stress response, chloroplast division, and nutrient signaling. There are three kinds of proteins responsible for the appropriate functioning of the TCS system: histidine kinases (HKs), histidine phosphotransfer proteins (HPs), and response regulators (RRs). The results of the current study revealed that *Vigna radiata* has 54 genes encoding potential TCS proteins, which were divided into three subgroups: 18 HKs, 9 HPs (seven true and two pseudos), and 27 RRs (8 type-A, 8 type-B, 3 type-C, and 8 PRRS). The anticipated TCS genes were widely dispersed across all eleven chromosomes and had family-specific intron/exon structures. After investigating TCS genes in a variety of plant species, we determined that *Vigna* HK (L)s, HPs, and RRs have closer evolutionary relationships with other legume genes. Gene duplication, including segmental and tandem types, is the most frequent source of gene family expansion. Multiple stress-related cis-elements were predicted in the promoter sequences of the VrTCS genes. RNA-seq data analysis demonstrated that VrTCS genes were expressed in clusters of upregulated and downregulated groups in response to drought stress. Moreover, these clusters were differentially expressed as early or late responses to drought stress. Real-time qPCR showed that *VrHK2*, *VrHK3*, *VrPHYE*, *VrHP4.1*, *VrRR5.2*, and *VrRR10* genes were upregulated, while *VrRR3* and *VrHP6.1* genes were downregulated in response to drought stress. The current study highlights the architecture of *V. radiata* TCS and provides a robust framework for subsequent functional evaluation.

Keywords: TCS; signaling; HKs; RRs; HPs; PPI; phylogenetic analysis



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1. Introduction

A crucial plant hormone commonly known as cytokinin is associated with several aspects of plant development, such as the formation of leaves and roots, seed germination, flowering, and the aging of plants [1–4]. Plants often transduce cytokinin signals through a process known as the two-component system [5]. Initially, this system was first reported in bacteria which was comprised of two signaling elements, namely, the histidine kinase (HK) gene family and the response regulator gene family [6]. The HK protein autophosphorylates when an external factor, such as a change in temperature, pH, antibiotics, or osmolarity supply of C, N, or PO₄, causes the phosphoryl group to be coupled with conserved histidine residues of the Hiska domain [7]. The activated HK protein subsequently transfers this

phosphoryl group to the RRs protein, where it binds with a conserved Asp residue of the Rec domain [8]. Over time, eukaryotes have acquired an intermediate protein family known as histidine phosphotransfer proteins as members of their multistep phosphorylation pathway. HP family members function as a linker protein in the signaling cascade, transferring the phosphate group from the histidine kinase protein to the response regulator protein [5,9].

Plants have three types of HKs: cytokinin receptors, phytochromes, and ethylene receptors. In *Arabidopsis thaliana*, three other HKs (*ACK1*, *CK12/AKH5*, and *AHK1*) do not belong to any particular class [9,10]. The general architecture of HKs consists of an input domain, a receiver (Rec) domain, a histidine-conserved transmitter domain (autophosphorylation site), and several N-terminus trans-membrane domains. Three ethylene receptors, *ERS2*, *AEIN4*, and *AETR2*, do not constitute autophosphorylation activity because phosphorylation-related residues are missing from their transmitter domains. As a result, they are characterized as diverging HKs [7,11]. In addition, cytokinin recognizes the cyclase/HK-associated sensory extracellular (CHASE) domain on the cytokinin receptors *AHK4*, *AHK3*, and *AHK2*. The phytochrome family includes the PHYE, PHYC, PHYB, PHYD, and PHYA members, as well as two PAS (Per/Arndt/Sim) folds and a chromophore-binding (PHY) domain [12,13].

The phosphotransfer (Hpt) domain of the HP family contains an evolutionarily conserved motif (XHQXKGSSXS) required for phosphoryl transfer from the HKs Rec domain to the RRs Rec domain [8,9,14]. AHP6 is known as a pseudo-HP protein because it lacks a His residue in its domain. AHP6 is also a cytokinin signaling suppressor that cannot function as a phosphotransfer protein [13,15]. Based on domain architecture, the response regulators were divided into three distinct categories: Type A, Type B, and Type C [7,16]. The majority of the type-A RRs are C-terminally extended cytokinin response proteins containing the Rec domain. Type B RRs belong to a class of transcriptional factors with an N-terminus Rec domain and a C-terminus output domain in their structure. The domain architecture of Type C RRs is similar to Type A RRs, but they are not activated by cytokinin, and no specific role of type-C RR in cytokinin signaling is recognized to date [17]. Moreover, there is a different class of RRs, generally called pseudo-RRs (PRRs). PRRs are not considered true RRs because they lack DDK-conserved motifs, in which essential residues for phosphorylation are absent. However, a specific motif known as CCT is present in the C-terminal extension, which is essential for regulating circadian rhythms [18–20].

Two-component system genes are linked to a wide range of abiotic stress tolerance, particularly responses to temperature, wind, water, and salt [21,22]. In plants, the TCS genes are involved in signal transduction, osmosensing, and important cellular processes such as ethylene, cytokinin, and red light responses [23,24]. Furthermore, the TCS genes are involved in activities such as nutrition sensing, stress response, chemotaxis, endosperm formation, and nodulation during plant growth, development, and adaptability [25,26].

Vigna radiata (L.) R. Wilczek var. *radiata* (Mung bean), also known as moong or green gram, is a staple food and a source of revenue in rice-based agricultural systems in Southeast and South Asia, though it is grown globally [27]. This crop is grown in both subsistence and commercial agriculture systems around the world for fiber, food (syrup and grain), fuel, and animal feed [28,29]. In most regions, mung bean yields are low, ranging from 0.5 to 1.5 t/ha. Temperature and salinity are two major factors that influence plant growth and development in a variety of ways [30,31]. As a result, crop productivity must be increased to meet global food demand in the coming decades. The current study sought to identify salt stress and drought stress-responsive TCS genes that may be beneficial in *V. radiata* development. Owing to the important role of the TCS genes in a wide range of biological activities, their identification and characterization must be carried out in *V. radiata*.

2. Materials and Methods

2.1. Gene Family Identification of Two-Component System in *V. radiata*

The Ensemble plants database was used to get *A. thaliana* TCS full-length protein sequences (<https://plants.ensembl.org/index.html>, accessed on 2 November 2021) [32].

The gene family members of TCS in *V. radiata* were identified by a BLASTp software (<https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE=Proteins>, accessed on 2 November 2021) search using *A. thaliana* sequences as a query [33]. The parameters of BLASTp were set as non-redundant protein sequence database, and BLOSUM62 was set as a scoring parameter. All TCS sequences with an E-value lower than 1×10^{-10} and the largest ORF were chosen for further investigation. Further, the TCS sequences that do not have a conserved domain for the functioning of TCS were eliminated using different domain databases, Pfam (<https://pfam.xfam.org/>, accessed on 20 November 2021) [34], ScanProsite (<https://prosite.expasy.org/scanprosite/>, accessed on 25 November 2021) [35], SMART (<http://smart.embl-heidelberg.de/>, accessed on 26 November 2021) [36], and CDD (<https://www.ncbi.nlm.nih.gov/Structure/cdd/cdd.shtml>, accessed on 2 December 2021) [37]. The ExPASy server's ProtParam tool was used to calculate the theoretical isoelectric point (pI), molecular weight (MW), grand average of hydrophobicity (GRAVY), aliphatic index, and instability index (<https://web.expasy.org/protparam/>, accessed on 10 December 2021) [38,39]. VrTCS gene subcellular localization was identified by using CELLO V.2.5 (<http://cello.life.nctu.edu.tw/>, accessed on 16 January 2022) [40]. The TCS genes' genomic information, such as genomic and cDNA sequences, chromosomal location, and protein length, was obtained from NCBI (<https://www.ncbi.nlm.nih.gov/>, accessed on 18 January 2022) [41]. The parameter of CELLO V.2.5 were set as protein fasta sequences and choose organism as eukaryotes.

2.2. Prediction of TCS Gene Structure and Cis Regulatory Elements in *V. radiata*

The schematic representation of the VrTCS gene structure (introns/exons) was carried out using the online software Gene Structure Display Service (GSDS) (<http://gsds.gao-lab.org/>, accessed on 24 January 2022) [42]. The input parameters were set as fasta sequence and output format were set as PNG format. NCBI was used to extract upstream 1000 bp genomic DNA sequences from the transcription start point of VrTCS genes. Then, these upstream sequences were subjected to the plant CARE website (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>, accessed on 26 January 2022) to identify potential cis-regulatory elements [43].

2.3. Phylogenetic Analysis, Multiple Sequence Alignment, and Motif Recognition

MEGA 7.0 software [44] was utilized to perform the phylogenetic analysis. For phylogenetic tree building, the Neighbor-joining (NJ) and Maximum Likelihood techniques were applied and additional factors such as bootstrapping (1000 replicates), poisson correction, and pairwise deletion, were considered. To construct phylogenetic relation TCS, the TCS protein sequences (HKs, Hpt, and RRs) of *A. thaliana*, *Glycine max*, *Sorghum bicolor*, *Cicer arietinum*, and *Oryza sativa* were retrieved from the database phytozome (<https://phytozome-next.jgi.doe.gov/>, accessed on 29 January 2022) [45]. Phylogenetic trees were built using these reported full-length protein sequences and the identified *V. radiata* potential TCS proteins. To evaluate the conservation of crucial residues involved in the phosphorylation process, ClustalX [46] (<http://www.clustal.org/>, accessed on 2 February 2022) was used to align sequences from several conserved domains (Rec, HK, Hpt, and Myb) and motifs (CCT). In addition, the MEME (Multiple EM for Motif Elicitation) program was used to anticipate the unique conserved motifs in the VrTCS protein sequence (<https://meme-suite.org/meme/>, accessed on 26 February 2022) [47]. The prediction range of the motif was fixed to twenty and the other thresholds were kept at their default values as classic mode for motif discovery, and the minimum width of the motif should be 6 and 60, respectively.

2.4. Chromosomal Mapping, Gene Duplication, and Evolutionary Analysis of TCS Members in *V. radiata*

A chromosomal genetic linkage map was created by using the advance Circos TBtool (<https://bio.tools/tbtools>, accessed on 18 March 2022) [48]. Furthermore, the location

of each TCS gene on *V. radiata* chromosomes was also determined using the NCBI-gene database. The gene duplication events were investigated using the DnaSP v.6 software (<https://bioinformaticshome.com/tools/descriptions/DnaSP.html>, accessed on 22 March 2022) [49]. To measure the selection pressure on the duplicated genes, the rates of synonymous and non-synonymous substitutions were determined. To trace evolutionary events, the divergence time was computed. $T Ks/2x$ ($x = 6.56 \times 10^{-9}$) formula was used to calculate the duplication time.

2.5. Two-Component System Gene Expression Patterns and Protein–Protein Interaction PPI in *V. radiata*

To further explore the expression level of these discovered *VrTCSs* in response to drought stress, RNA-seq data on seed (BioProject: PRJNA327304) was downloaded from the SRA database (<https://www.ncbi.nlm.nih.gov/sra>, accessed on 29 March 2022). From NCBI (https://www.ncbi.nlm.nih.gov/assembly/GCF_000741045.1/, accessed on 29 March 2022), the genome annotation files (.gtf and .fna) were downloaded. Bowtie2 was used to create indexes of the *V. radiata* genome sequence, and paired-end clean reads of high quality were mapped to the *V. radiata* genome. The cufflinks software was then utilized to compute the expression levels of the reference genome's annotated genes. The normalized FPKM values of each *VrTCS* were computed to determine whether genes were differently expressed, either up-regulated or down-regulated [50]. The statistical analysis and measures for RNA sequence analysis and differential gene expression were same as reported in BioProject: PRJNA327304. TBtool was used to create a heatmap of the expression. To further understand how these TCS proteins respond to abiotic stress, we projected a protein interaction network for these TCS proteins. The *VrTCS* protein sequences were input into the STRING database to create the protein interaction [51]. In PPI, proteins having an interaction score > 0.700 were selected for further analysis.

2.6. Plant Growth and Treatments

V. radiata plants were cultivated in a growth chamber for 28 days under the following conditions. The temperature ranged from 25 to 27 °C throughout the experimental duration, whereas the light/dark cycle was for 16/8 h. The relative humidity level was maintained at 65%. Twenty-eight days after the cultivation, the drought stress was applied. For the drought treatment, the watering to the selected pots was stopped. The pots in control were watered to normal field capacity, i.e., 80–100%. Leaf samples were collected from each pot (control, and drought at day 0, day 3 and day 5 with three biological replicates for RNA extraction.

2.7. Validation of Expression Results Using Quantitative Real-Time PCR

Leaf samples were crushed using antiseptic pestles and mortar in the presence of liquid nitrogen. The Fastlane cell cDNA kit (Qiagen, Switzerland) was used to synthesize complementary DNA (cDNA). The RNA was quantified using a Nanodrop spectrophotometer (NanoDrop 2000, Thermo Fisher Scientific Waltham, MA, USA). One microgram of the RNA sample was used for cDNA synthesis with an All-in-One First-Strand synthesis kit (Thermo Fisher Scientific, Waltham, MA, USA). The SYBR Green Master kit was used to perform the qPCR reactions in the Applied Biosystem Real-Time PCR Detection System (RT-PCR). TCS gene-specific primers (Supplementary Table S1) were constructed using the “Oligo Calculator” online tool (<http://mcb.berkeley.edu/labs/krantz/tools/oligocalc.html>, accessed on 18 June 2022) [52], and their specificity was confirmed using the NCBI Primer-BLAST program (<https://www.ncbi.nlm.nih.gov/tools/primer-blast/>, accessed on 18 June 2022) [53]. To standardize gene expression, the reference gene glyceraldehyde-3-phosphate dehydrogenase (GAPDH) was employed [54,55].

3. Results

3.1. Comprehensive Identification of Two-Component System Genes in *V. radiata*

The current study used a BLASTp search to identify putative TCS gene family members in *V. radiata* using the TCS protein sequences of *A. thaliana*. *V. radiata* genome contained 54 TCS genes, which were further classified as 18 HK, 9 HP, and 27 RR.

3.2. Histidine Kinase Protein Family in *V. radiata*

In the current genome-wide analysis, the genome of *V. radiata* contains 18 HKs (*VrHKs/VrHKLs* (histidine kinase like)) sequences which are more than the HKs sequences of *O. sativa* (5), *Z. mays* (11), *S. bicolor* (13), *T. aestivum* (7) and *P. trichocarpa* (12). However, this higher proportion of HKs is close to other plants such as *C. melo* L. (17), *A. thaliana* (17), *G. max* (21), *B. rapa* (20), *S. lycopersicum* (18) and *L. japonicus* (14), indicating that HKs may be important in these species (Table 1). *VrHK1.1*, *VrHK1.2*, *VrHK2*, *VrHK3*, *VrHK4.1*, *VrHK4.2*, *VrHK5*, *VrCK1.1*, and *VrCK1.2* were discovered to be cytokinin-related genes in *V. radiata*. The conserved residues required for HK activity appeared to be present in all members (Figure 1; Supplementary Table S2; Figure S1). All of these HKs have a conserved HisKa domain with a conserved His phosphorylation site, according to domain analysis. Furthermore, each of the nine members possessed a conserved RR (Rec) domain containing a highly conserved Asp residue that functions as a phosphoreceptor. According to genetic and molecular studies, the ethylene response in *A. thaliana* is mediated by five ethylene receptors. These receptors contain a GAF protein–protein interaction domain, a HisKa domain, and a HATPase domain [56]. Similarly, six genes in *V. radiata* were predicted to code ethylene receptors (*VrETR1*, *VrETR2*, *VrERS1*, *VrERS2*, *VrEIN4.1*, and *VrEIN4.2*). *VrETR1*, *VrERS1*, and *VrERS2* have domains that are similar to those found in *A. thaliana*. While *VrETR2*, *VrEIN4.1*, and *VrEIN4.2* contain GAF, HisKa, and a conserved Rec domain (Figure 1; Supplementary Table S2, Figure S1).

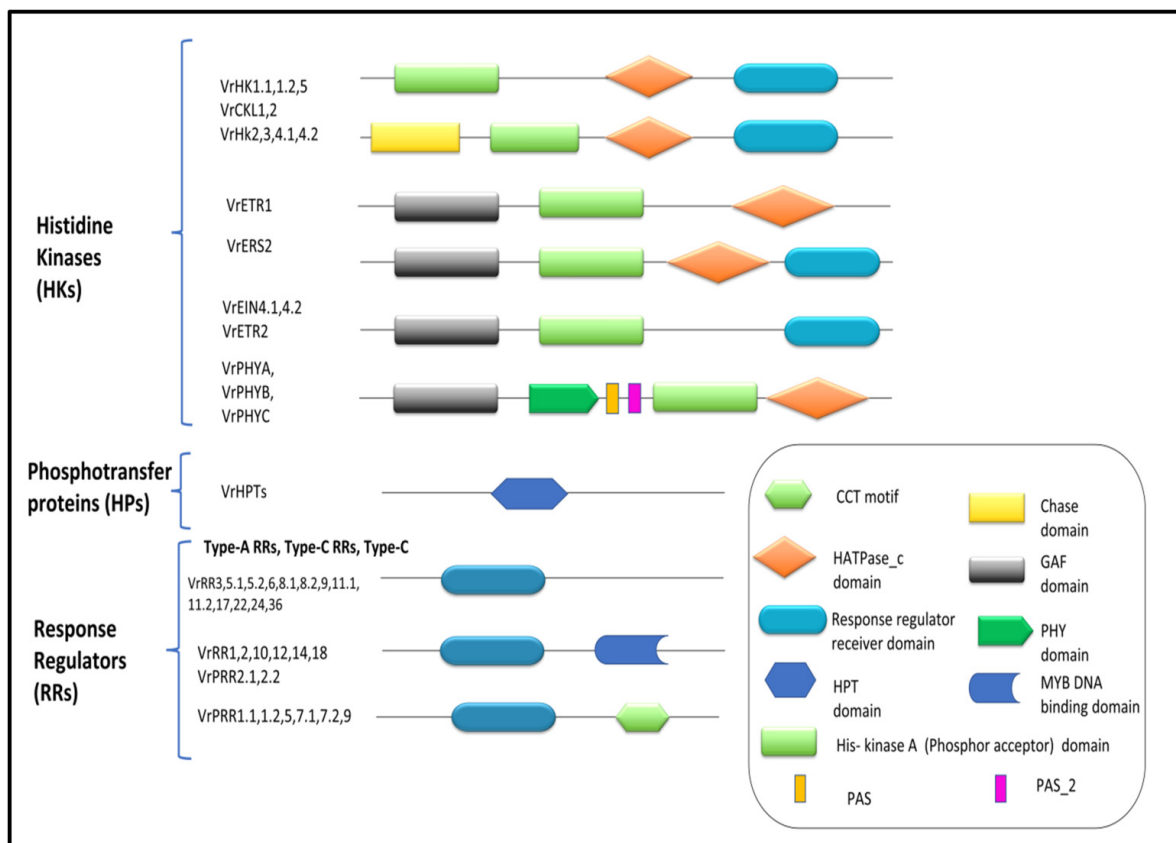


Figure 1. Domain architecture of *V. radiata* TCS proteins. Different colored shapes represent different domains.

The developmental and growth processes of plants in response to light stimuli are mediated by photoreceptors or the phytochrome family. *PHYA*, *PHYB*, *PHYC*, *PHYD*, and *PHYE* have been identified as phytochrome families in *A. thaliana* [57]. Phytochrome consists of an N-terminal PHY domain that is responsible for the absorption of light. One HisKa, GAF domain, and two PAS domains are involved in signal transduction. The structures of phytochrome are similar to those of sensor H proteins and are soluble proteins. The signal transduction (HisKa) domain is located at the C-terminal and the sensor domain is located at the N-terminal site. The phytochrome family is also known as divergent HKs because they lack all the conserved motifs. They contain Ser/Thr kinase activity instead of HKs activity for signal transduction [58]. In the genome of *V. radiata*, three members *VrPHYA*, *VrPHYB*, and *VrPHYE* were found. All three members contain PAS, GAF, PHY, and HisKa domains that are responsible for light and signal transduction. The presence of these domains makes them true photoreceptors (Figure 1; Supplementary Table S2, Figure S2).

3.3. Histidine Phosphotransfer Protein (HpT) Family in *V. radiata*

Six members of the HP gene family (*AHP1*–*AHP6*) have been discovered in the genome of *A. thaliana*, five of which are true HPs and one of which is a pseudo-HP because it lacks a conserved histidine (H) residue required to accept the phosphate group from the donor protein [56,59]. The histidine (H) residue in *AHP6/APHP1* was replaced by an (N) residue, making this member a pseudo-HP, whereas a conserved phosphorylation motif (XHQXKGSXS) was present in all other members of the HP family. In the genome of *V. radiata*, nine potential HP gene family members (*HP1.1*, *HP1.2*, *HP1.3*, *HP2*, *HP4.1*, *HP4.2*, *HP5*, *HP6.1*, and *HP6.2*) were discovered. *VrHP6.1* and *VrHP6.2* genes were identified as pseudo-HPs due to the absence of phosphorylating conserved residue (Figure 1; Supplementary Table S2, Figure S2).

3.4. The Response Regulators (RR) Family in *V. radiata*

The final response to the environmental stimulus was regulated by the response regulator (RR). In the *A. thaliana* genome, 32 putative RR family members were discovered, and 22 in *O. sativa* [9,60]. In this study, a total of 27 putative response regulators (RRs) were discovered in the *V. radiata* genome. In the signaling pathway of TCS, response regulators act as terminal components that perform a function as phosphorylation-activated switches. The RR family were divided into three subfamilies based on conserved domains: the first one is type-A, the second is type-B and the third is type-C RRs. Type-A RRs have a conserved aspartic acid (D) residue in their Rec domain and a long C-terminus extension. Type-B RRs have a conserved Rec domain and a Myb binding domain, whereas type-C RRs have a domain structure like type-A RRs but lack the C-terminal extension. Another type of RR is the pseudo-response regulator, which has a conserved Rec domain in which the conserved aspartic acid (D) residue is replaced by glutamic acid (E) and a C-terminal CCT motif. A comprehensive genome-wide analysis of *V. radiata* reveals the presence of 27 RR family-related members. There were eight type-A, eight type-B, three type-C, and eight PRRs present. The type-A RR family in *V. radiata* is made up of eight members: *VrRR3*, *VrRR5.1*, *VrRR5.2*, *VrRR6*, *VrRR8.1*, *VrRR8.2*, *VrRR9*, and *VrRR17*, all of which are likely to be identical to their *A. thaliana* counterparts and have a conserved Rec domain [18].

Because they contain a Myb-DNA binding domain, type-B RRs were discovered to be involved in transcriptional regulation. *VrRR1*, *VrRR2*, *VrRR10*, *VrRR11.1*, *VrRR11.2*, *VrRR12*, *VrRR14*, and *VrRR18* are members of this family in *V. radiata*, which is more than *O. sativa* (7), *C. arietinum* (7), *T. aestivum* (2), and *S. bicolor* (7), but less than *A. thaliana* and *G. max* [4,58–60]. Except for *VrRR11.1* and *VrRR11.2*, which are missing the Myb domain, all nine members have conserved Myb and Rec domains. *V. radiata* has three members of the type-C RR family (*VrRR22*, *VrRR24*, and *VrRR36*), while *A. thaliana* has two (*ARR22* and *ARR24*) [18]. These members share a Rec domain with their *A. thaliana* counterparts and have strong phylogenetic homologous relationships. The *V. radiata* genome contains eight

PRRs: *VRPRR1.1*, *VRPRR1.2*, *VRPRR2.1*, *VRPRR2.2*, *VRPRR5*, *VRPRR7.1*, *VRPRR7.2*, and *VRPRR9*. PRRs were further classified into two types based on their C-terminal extension: clock-associated PRRs (which contain the CCT motif) and type-B PRRs (contain the Myb domain). *VrPRR2.1* and *VrPRR2.2* are the only two genes with a Myb domain rather than a CCT motif, indicating that they are type-B PRRs (Figure 1; Supplementary Table S2, Figure S3).

Table 1. TCS genes reported in a variety of plants.

Species	Total No. of TCS Genes	HK	HP (Pseudo HP)	Type-A RRs	Type B RRs	Type C RRs	PRRS	Cotyledons	References
<i>Arabidopsis thaliana</i>	47	8	6 (1)	10	12	2	9	Dicot	[15]
<i>Oryza sativa</i> L.	37	5	5	15	7	0	5	Monocot	[60]
<i>Cicer arietinum</i>	51	18	7 (2)	7	7	2	10 b	Dicot	[58]
<i>Sorghum bicolor</i>	37	13	5 (2)	3	7	2	7 b	Monocot	[59]
<i>Glycine max</i>	98	36	13	18	15	3	13	Dicot	[10]
<i>Citrullus lanatus</i>	49	19	6 (2)	8	10	6	5	Dicot	[61]
<i>Cucumis melo</i> L.	51	17	9	8	11	0	6	Eudicots	[62]
<i>Cucumis sativus</i> L.	46	18	7 (2)	8	8	0	5	Dicot	[61]
<i>Populus trichocarpa</i>	49	12	12	9	11	0	5	Dicot	[63]
<i>Solanum lycopersicum</i>	65	20	6 (2)	7	23	1	8	Dicot	[64]
<i>Physcomitrella patens</i>	39	18	3	7	5	2	4 a	Monocot	[65]
<i>Zea mays</i>	59	11	9 (2)	16	9	3	11 a	Monocot	[66]
<i>Zizania latifolia</i>	69	25	8	14	14	2	6	Monocot	[67]
<i>Lotus japonicus</i>	40	14	7	7	11	1	5 a	Dicot	[20]
<i>Brassica rapa</i>	85	20	8 (1)	21	17	4	15	Dicot	[68]
<i>Triticum aestivum</i>	62	7	10	41	2	0	2	Monocot	[69]
<i>V. radiata</i>	54	18	9(2)	8	8	3	8	Dicot	Present work

a Clock-associated. b Both type-B PRRs and clock-associated.

3.5. Features of VrTCS

The genomic Table 2 examines the detailed physio-chemical properties of the 54 VrTCS proteins. VrTCSs are found on chromosomes 1–10. The length of the proteins ranged from 111 amino acids (*VrHP5*) to 1265 amino acids (*VrHk1.1*), and the number of exons ranged from 2 to 14. ExPASy analysis shows that the VrTCS proteins' isoelectric points range from 4.53 to 8.69, their molecular weights range from 12,940.53 Da to 141,438.69 Da, and their stability index ranges from 24.28 to 71.26. In addition, the aliphatic index and GRAVY of VrTCS were computed, with values ranging from 66.68 to 107.17 and 0.828 to 0.122, respectively.

Table 2. Genomic features of *VrTCS*.

Locus ID	Gene Name	Isoelectric Point (PI)	Molecular Weight (Da)	Instability Index	Aliphatic Index	GRAVY	Protein Length (Amino Acids)	Chr No.	No. of Exon	G Start	G End	Cellular Localization
HK Family												
LOC106755439	<i>VrHK3</i>	6.7	114,682	37.61	93.55	−0.107	1028	Un	10	110,644	118,425	Plasma Membrane
LOC106764676	<i>VrHK2</i>	7.2	134,098	43.76	87.55	−0.22	1190	C6	14	479,116	488,795	Plasma Membrane
LOC106777844	<i>VrHK4.1</i>	6.67	110,950	36.19	89.42	−0.124	990	C11	14	16,816,322	1,683,310	Cytoplasmic
LOC106770782	<i>VrHK4.2</i>	6.82	111,430	33.66	86.49	−0.174	999	C1	12	4,438,961	4,447,994	Cytoplasmic
LOC106770468	<i>VrCK11</i>	6.82	111,430	33.98	90.75	−0.115	1060	C8	7	28,809,958	28,814,950	Plasma Membrane
LOC106777758	<i>VrHK1.1</i>	6.18	141,439	41.26	90.86	−0.223	1265	C11	13	6,536,759	6,543,727	Plasma Membrane
LOC106768916	<i>VrHK1.2</i>	8.31	137,838	40.22	89.76	−0.253	1234	C7	13	2,505,103	2,511,396	Plasma Membrane
LOC106761316	<i>VrCK12</i>	4.99	109,374	56.97	82.21	−0.445	972	C5	13	20,096,528	20,101,731	Nuclear
LOC106763060	<i>VrHK5</i>	5.15	114,127	54.95	77.67	−0.567	1011	C6	13	315,594,338	31,567,102	Nuclear
LOC106756118	<i>VrETR1</i>	8.31	83,132.2	40.46	107.2	0.122	742	C2	7	24,449,047	24,457,030	Plasma Membrane
LOC106757732	<i>VrERS1</i>	6.36	70,848.1	41.01	102.4	0.031	636	C3	7	7,678,537	7,683,030	Plasma Membrane
LOC106754157	<i>VrEIN4.1</i>	6.91	85,246.9	35.96	101	0.053	760	Un	2	1,092,832	1,096,427	Plasma Membrane
LOC106754157	<i>VrEIN4.2</i>	6.65	85,181.1	36.55	103.3	0.086	759	Un	2	1,098,989	1,102,570	Plasma Membrane
LOC106772158	<i>VrERS2</i>	8.05	83,779.4	41.93	103.9	0.085	757	C8	5	35,402,755	35,407,049	Plasma Membrane
LOC106772500	<i>VrETR2</i>	6.28	84,271.9	33.78	97.53	0.111	760	C8	3	1,031,509	1,035,564	Plasma Membrane
LOC106771983	<i>VrPHYA</i>	6.09	123,961	47.23	93.94	−0.079	1123	C8	7	31,264,701	31,270,514	Cytoplasmic
LOC106760765	<i>VrPHYE</i>	5.69	125,312	46.74	90.79	−0.173	1121	C5	4	4,573,654	4,579,012	Cytoplasmic, Plasma Membrane,
LOC106754114	<i>VrPHYB</i>	5.69	125,677	45.85	93.98	−0.116	1131	Un	4	10,541	17,443	Cytoplasmic

Table 2. Cont.

Locus ID	Gene Name	Isoelectric Point (PI)	Molecular Weight (Da)	Instability Index	Aliphatic Index	GRAVY	Protein Length (Amino Acids)	Chr No.	No. of Exon	G Start	G End	Cellular Localization
HPTs												
LOC106775067	VrHP1.1	4.8	17,631	42.53	88.65	−0.254	155	C10	7	18,301,338	16,303,340	Cytoplasmic, Extracellular
LOC106767217	VrHP1.2	4.74	17,736.2	43.93	90.45	−0.223	154	C7	6	49,140,073	49,141,539	Cytoplasmic, Nuclear
LOC106771462	VrHP1.3	5.03	16,565.7	37.7	94.73	−0.34	146	C8	6	32,351,139	32,352,533	Nuclear
LOC106771501	VrHP2	5.45	17,067.4	43.33	74.8	−0.361	150	C8	6	26,939,465	26,942,841	Nuclear
LOC106773913	VrHP4.1	8.68	17,413.7	62.07	71.85	−0.746	151	C9	6	1,290,483	1,292,013	Nuclear
LOC106761115	VrHP4.2	8.59	23,040.5	36.03	79.29	−0.365	198	C5	6	31,782,312	31,785,137	Nuclear
LOC106760377	VrHP5	4.53	12,940.5	49.06	70.27	−0.48	111	C5	4	16,465,489	16,467,894	Nuclear, Cytoplasmic
LOC106773969	VrHP6.1	5.94	17,971.5	44.3	90.71	−0.271	156	C9	6	547,257	548,917	Extracellular
LOC106761820	VrHP6.2	7.59	25,968.9	48.28	89.69	−0.116	223	C5	5	25,414,244	25,415,627	Extracellular, Plasma Membrane
Type-A RRs												
LOC106760963	VrRR3	4.81	25,479.8	66.93	88.06	−0.309	232	C5	5	7,566,240	7,568,017	Nuclear
LOC106766897	VrRR5.1	7.63	22,881.7	55.32	103.1	−0.143	206	C7	5	35,245,726	35,247,771	Nuclear
LOC106758083	VrRR6	7.7	21,905.2	57.57	91.09	−0.116	202	C4	5	15,357,967	15,359,998	Nuclear, Chloroplast
LOC106767155	VrRR5.2	6.76	22,762.5	55.51	102.1	−0.121	206	C7	5	35,254,959	35,257,043	Nuclear, Cytoplasmic
LOC106752532	VrRR9	5.69	27,500.9	71.26	74.98	−0.828	243	Un	5	359,726	361,726	Nuclear
LOC106779256	VrRR8.1	5.72	20,486.6	59.82	88.72	−0.401	179	Un	5	344,091	345,589	Nuclear
LOC106760951	VrRR8.2	4.95	24,924.7	68.98	93.93	−0.276	219	C5	4	26,430,472	26,431,946	Nuclear, Plasma Membrane
LOC106759802	VrRR17	8.69	15,785.5	47.54	94.69	−0.208	143	C5	5	9,332,686	9,333,984	Nuclear, Extracellular

Table 2. Cont.

Locus ID	Gene Name	Isoelectric Point (PI)	Molecular Weight (Da)	Instability Index	Aliphatic Index	GRAVY	Protein Length (Amino Acids)	Chr No.	No. of Exon	G Start	G End	Cellular Localization
Type-B RRs												
LOC106768205	VrRR1	5.84	77,305.4	45.05	82.15	−0.393	706	C7	7	48,240,050	48,244,152	Nuclear
LOC106760138	VrRR2	6.07	73,630.9	46.03	81.08	−0.441	673	C5	6	734,320	738,195	Nuclear
LOC106777889	VrRR14	6.15	68,215.7	56.73	74.84	−0.49	620	C11	7	17,020,135	17,027,339	Nuclear
LOC106762533	VrRR18	6.21	73,998.2	41.95	77.94	−0.562	670	C5	6	18,903,242	18,907,979	Nuclear
LOC106757890	VrRR12	5.44	73,702.2	48.21	76.48	−0.516	677	C3	6	11,915,194	11,920,109	Nuclear
LOC106763713	VrRR10	6.4	67,500.7	41.87	76.44	−0.504	613	C6	9	32,044,562	32,048,926	Nuclear
LOC106777203	VrRR11.1	5.57	64,930	48.37	76.71	−0.527	578	C2	6	22,436,393	22,440,444	Nuclear
LOC106766713	VrRR11.2	5.06	65,044.1	56.94	76.23	−0.503	578	C7	5	20,462,824	20,467,554	Nuclear
Type-C RRs												
LOC106754035	VrRR22	6.9	15,514.9	24.28	72.1	−0.395	138	Un	2	964,858	965,390	Cytoplasmic
LOC106757536	VrRR36	8.49	16,082.7	26.09	84.48	−0.275	143	C3	3	3,714,256	3,715,525	Chloroplast, Mitochondrial
LOC106756903	VrRR24	7.94	14,994.3	47.14	88.65	−0.382	133	C3	2	3,954,702	3,955,257	Nuclear, Mitochondrial
PRRs												
LOC106769035	VrPRR1.2	5.67	63,861.3	53.14	73.33	−0.642	570	C7	7	38,576,824	38,582,446	Nuclear
LOC106756669	VrPRR1.1	5.95	62,408.7	57.49	66.68	−0.724	557	C3	6	10,571,692	10,576,727	Nuclear
LOC106777717	VrPRR2.1	5.89	62,525.5	56.81	67.57	−0.668	560	C11	13	3,330,688	3,336,996	Nuclear
LOC106769136	VrPRR2.2	6.22	62,034.5	54	71.81	−0.573	554	C7	13	29,778,486	29,785,007	Nuclear
LOC106754416	VrPRR7.1	7.54	80,057.9	43.73	68.81	−0.734	737	Un	9	279,157	313,750	Nuclear
LOC106760140	VrPRR5	6.5	72,478.7	42.36	66.99	−0.656	655	C5	8	404,026	408,131	Nuclear
LOC106780351	VrPRR9	6.05	74,267.3	44.64	67.32	−0.656	675	Un	10	15,087	20,882	Nuclear
LOC106754416	VrPRR7.2	7.54	81,386.4	43.67	69.52	−0.721	749	Un	9	279,157	313,750	Nuclear

3.6. Gene Structure and Conserved Motif Analysis of *V. radiata* TCS Members

A gene's intron-exon structure is an important evolutionary feature that provides insight into its functional variation. As a result, the intron-exon arrangements of the VrTCS genes were investigated. The cytokinin receptor HK family has exons ranging from 10 in *VrHK3* to 14 in *VrHK4.2*, with introns ranging from 9 to 13. *VrCK1* has six introns and seven exons in its gene sequence, whereas *VrCK2* has twelve introns and thirteen exons. TCS members *VrEIN4.1* and *VrEIN4.2* had two exons and three introns, whereas *VrERS1*, *VrETR1*, *VrETR2*, and *VrERS2* have seven, seven, three, and five exons in their gene structures, respectively. *VrPHYA* has seven exons, while *VrPHYB* and *VrPHYE* have four exons each.

The gene structure of the HP family consists of four to seven exons. Exon counts in the *V. radiata* RR family genes ranged from 2 to 13. The most exons were discovered in *VrPRR2.1* and *VrPRR2.2*. After that, we used the MEME program to predict the VrTCS conserved motifs. MEME software discovered a total of 20 motifs, of which motifs 8, 10, 11, and 18 were conserved throughout the HK family. The ethylene receptor family also contained similar motifs, with additional conserved motifs 9, 12, 14, and 19. There are between eight and ten non-conserved motifs in the phytochrome family; seven and nine were conserved in the HP family motifs. Overall, Type A RRs shared motifs 1, 2, 4, and 13, whereas Type C shares all but motif 2. Type-B has a motif similar to type-A, with one additional conserved motif 20, and motif 15 is present in PRRs (Figure 2, Supplementary Figure S4). Members of the same gene families share motifs, indicating that there is no sequence divergence between them.

3.7. Phylogenetic Analysis of *V. radiata* TCS Members

We further investigated the evolutionary trend and phylogenetic relationships of TCS proteins in *V. radiata*. The neighbor-joining tree was constructed using TCS peptide sequences for *V. radiata*, *A. thaliana*, *C. arvensis*, *G. max*, *S. bicolor*, and *O. sativa* (Figure 3) [58–60]. These TCS proteins were classified into three groups: phosphotransfer proteins (HPs), histidine kinases (HKs), and response regulators (RRs). The histidine kinase subunits were classified into five groups in the evolutionary tree: phytochromes, HK1, CK1, cytokinin receptor, CK2, and ethylene receptor. It has been revealed that both *A. thaliana* and *V. radiata* contain cytokinin receptors. They were identified in *A. thaliana* as *AKH4*, *AKH3*, and *AKH2*, and in *V. radiata* as *VrHK2*, *VrHK3*, *VrHK4.1*, and *VrHK4.2*. The HKs tree revealed that *VrHK3* and *VrHK4* were orthologues of the AHKs. The functional analysis of *A. thaliana* cytokinin receptors revealed that they are involved in a wide range of cytokinin activities, including cell differentiation, vascular differentiation, stress responses, leaf senescence, mitosis, and seed size. Based on the formation of cytokinin groups in *V. radiata* and *A. thaliana* in the HK tree, it is possible that *V. radiata*'s cytokinin serves a similar function.

AHK1 is an osmosensing transmembrane protein in *A. thaliana* that is primarily expressed in the roots under salt-stress conditions. *VrHK1.1*, *VrHK1.2*, and *VrCKL1* are orthologues of AHK1 in *V. radiata* [70]. *VrHK5* and *VrCK2* are the orthologs of AHK5/ACK2 in the CK2 group, according to phylogenetic analysis. *VrETR1* and *VrERS1* are ethylene receptors that are direct homologs of *ETR1* and *ERS1*. *VrEIN4.1* and *VrEIN4.2* proteins are also found on another branch, forming a direct ortholog link with *ERS2* and *ETR2* in *A. thaliana* (Figure 3; Supplementary Figure S5) [59].

Photoreceptors known as phytochromes have been discovered to be important in plant development and growth under light-stress conditions [71]. *A. thaliana* has five photoreceptors, including one HisKa, two PAS domains, a GAF domain, and a C-terminal, N-terminus PHY domain, all of which are important in the transduction of signaling pathways [72]. Three photoreceptors, *VrPHYA*, *VrPHYB*, and *VrPHYE*, were discovered in *V. radiata* with the same domain as those found in *A. thaliana*. All the putative identified VrPHYs were direct homologs of AtPHYs. *V. radiata* contains nine HPs that are all closely related to the true HPs found in *O. sativa* and *A. thaliana*. These HPs were classified based

on their evolutionary relationships with *A. thaliana* counterparts. *VrHP1.1*, *VrHP1.2*, and *VrHP1.3* were identified as AHP1-like. *VrHP2* was grouped as AHP2-like, while *VrHP6.1* and *VrHP6.2* were grouped as being AHP6-like, whereas *VrHP4.1*, *VrHP4.2*, and *VrHP5* were categorized with AHP4-like HPs (Figure 3; Supplementary Figure S6).

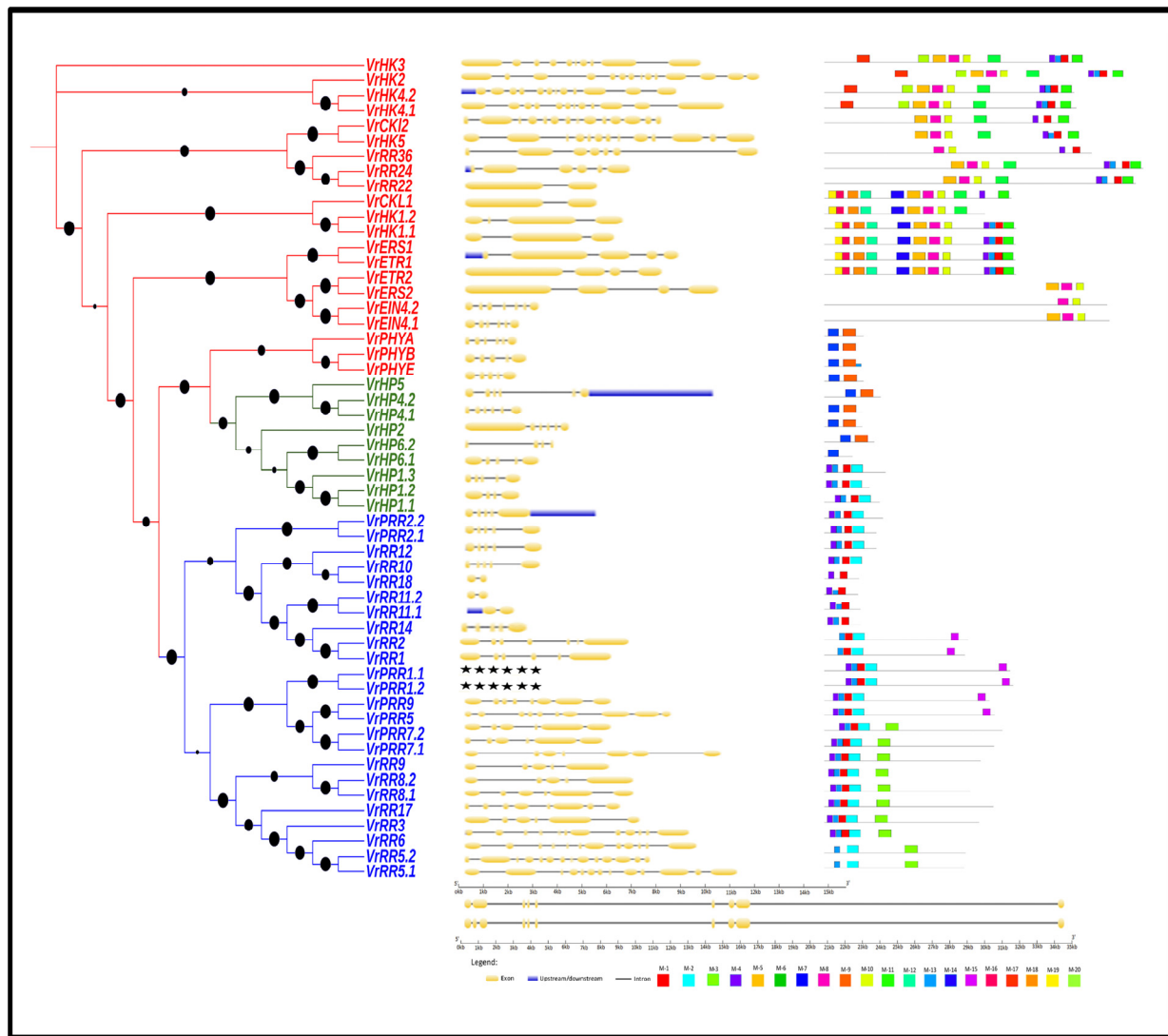


Figure 2. Representation of *V. radiata* TCS gene family's conserved motif and gene structure. In every gene structure, the yellow bar represents the exon, while the black line represents the intron part of the corresponding gene, and the blue bars show the untranslated region (UTR). In conserved motifs, distinct colors represent different motifs.

For the evolutionary study of response regulators, the sequence RRs proteins of *A. thaliana*, *V. radiata*, *O. sativa*, *C. arifinum*, *S. bicolor*, and *G. max* were utilized [58–60]. The third partner in the signaling TCS transduction pathways that trigger the genes in response to stress is response regulators, which are classified as Type A, Type B, Type C, and PRRs. All type-A RRs have a close relationship to their *A. thaliana* and *O. sativa* counterparts, indicating that they are genuine type-A RRs. The evolution pattern of *V. radiata* revealed that segmental duplication occurs in their genome. Similar findings have been obtained in *A. thaliana* and *O. sativa*, both of which exhibit segmental duplication. Type-A RRs are considered novel members of the RR family because they are found only in terrestrial plants and are absent in unicellular algae. They have been proposed to engage in a variety of novel activities in those species. The attachment of the phosphate group in their structure

requires a conserved DDK and Rec domain motif [73]. Cytokinin primarily activates type-A RRs, with type-B RRs also contributing to cytokinin induction (Figure 3; Supplementary Figure S7).

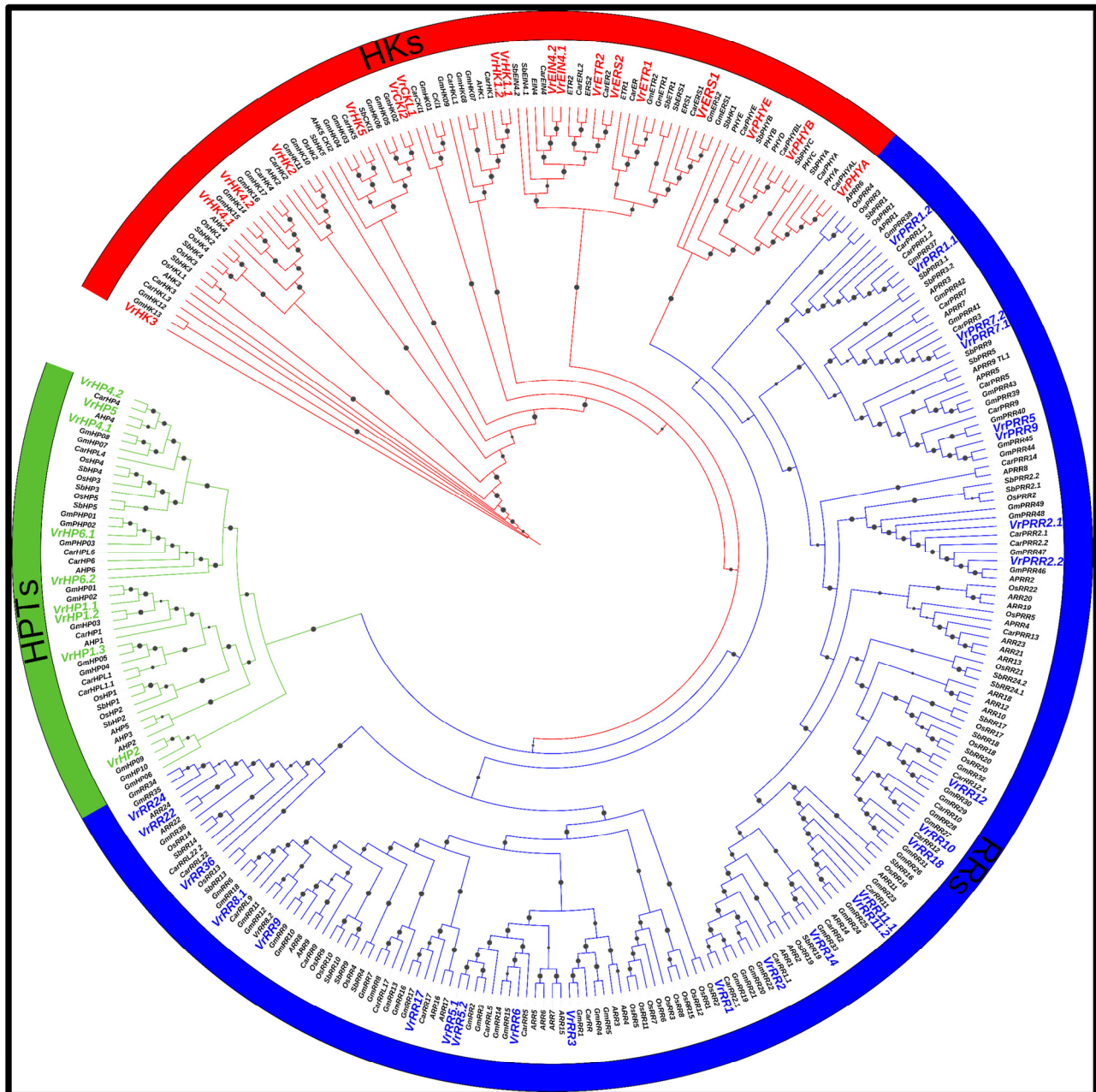


Figure 3. Evolutionary analysis of VrTCS with other plant species. The red color clades represented histidine kinase, blue represents the response regulators and the histidine phosphotransfer is represented by green colors.

3.8. Synteny Analysis, Gene Distribution, and Duplication of *V. radiata* TCS Members

The VrTCS gene distribution, as well as the gene location on the chromosome and gene duplicated pairs, were determined using Synteny analysis. All of the TCS family members identified in *V. radiata*—except for a few genes, due to a lack of locus information—are distributed across eleven chromosomes. The distribution of genes on chromosomes is unequal, with chromosome 5 containing eleven genes (the maximum number of genes present), chromosome 7 containing eight genes, and chromosomes 1, 4, and 10 containing

only one gene. Except for chr4, chr9, and chr10, the HKL-related genes are dispersed randomly across all *V. radiata* chromosomes. Members of the TCS HP family can be found on chromosomes 5, 7, 9, and 10. Apart from chromosomes 1, 8, 9, and 10, members of the TCS RR family are found on all other chromosomes (Figure 4).

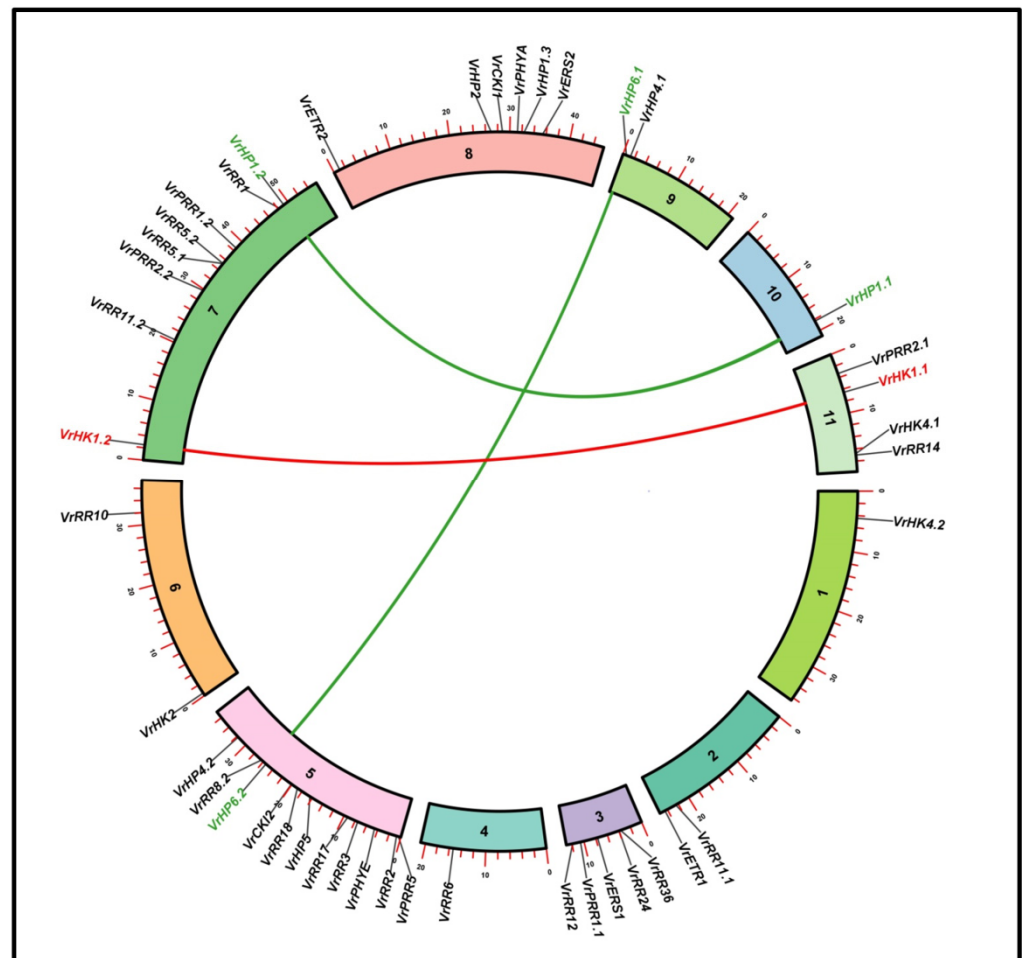


Figure 4. Representation of potential *VrTCS* genes on chromosomes. The circular bar scale indicates the length of the chromosomes in MBs and the numbers in the circular bar represent chromosomes number. The duplicated gene pair is represented by colored lines.

Gene duplication is dominant in the evolution of gene families because it provides the raw resources for the introduction of unique genes. The gene pool of plants has grown as a result of tandem or segmental gene duplication [74]. When investigating the putative genomic duplication events, three possible pairs of paralogs were discovered in the genome of *V. radiata*. In this study, *VrHHP1.1/VrHHP1.2*, *VrHHP6.1/VrHHP6.2*, and *VrHHP18/VrHHP17* were among the duplicated pairs, discovered as a consequence of segmental duplication. Multiple pairs of *V. radiata* genes showed segmental duplication, indicating that segmental duplication is the primary cause of *VrTCS* gene expansion. Different plants, such as *A. thaliana*, *S. bicolor*, Chinese cabbage, and *G. max*, also showed a similar pattern of gene expansion [10,59,68].

The synonymous/nonsynonymous substitution rates and the K_a/K_s ratio were used to estimate the gene divergence mechanism. K_s and K_a modes for segmental duplication were computed to reflect the time of paralogous divergence and evolutionary trend in *V. radiata*. Three segment duplicates had K_a/K_s ratios ranging from 0.918083957 to 2.395882445. As a result, the divergent time spans 79.59 to 193.99 Mya (Table 3).

Table 3. Divergence time and duplicated gene pairs in VrTCS.

Gene Duplication Pairs	Ka	Ks	Ka/Ks	Divergence Time Mya (Million Years Ago)	Type of Duplication
<i>VrHK1/VrHK1.2</i>	3.463	1.585	2.185	120.80	Segmental
<i>VrHP1.1/VrHP1.2</i>	6.098	2.545	2.396	193.99	Segmental
<i>VrHP6.1/VrHP6.2</i>	2.294	2.499	0.9180	190.46	Segmental

3.9. Promoter Analysis of *V. radiata* TCS Genes

The upstream sequences of the VrTCS genes were investigated in order to predict the cis-regulatory elements to gain a better understanding of their regulating expression and functional significance. In the upstream sequences of TCS genes, several hormone-related (gibberellins responsiveness, auxin, ethylene, MeJA responsiveness, abscisic acid responsiveness, and salicylic acid responsive) and abiotic stress-related (cold, drought, high temperatures, light, and high salinity) cis-regulatory elements were predicted [75]. All VrTCSs contained a large number of light responsiveness-related cis-regulatory elements (G-box, GATA motif, TCCC motif, ACE, box 4, and TCT motifs). It was discovered that nearly 20 VrTCSs genes contain gibberellin-responsive components (TATC-box, GARE motif, and P-box) and 33 VrTCSs genes contain an abscisic acid-responsive element (ABRE) that is associated with abscisic acid response. Elements of low-temperature responsive elements (LTR), stress and defense responsive elements, salicylic acid-responsive elements, and MeJA-responsive elements were also discovered in 4, 12, 12, and 24 VrTCSs, respectively. The occurrence of MBS (MYB binding site), TC-rich repeats (defense responsive element), and LTRs, all of which are associated with drought-inducibility, indicates that TCS is crucial for the growth of the plant and in response to abiotic stress [76]. TCS genes may potentially participate in development and growth processes involving hormone metabolism and signal transduction networks according to these findings (Figure 5; Supplementary Table S3; Figure S8).

3.10. Expression Analysis of *V. radiata* TCS Genes

The SRA-NCBI database was used to collect publicly available RNA-seq data from *V. radiata* in order to assess the expression of the 54 VrTCS genes under abiotic stress conditions. Drought tolerance studies in *V. radiata* were carried out across a wide range of time frames to collect data on abiotic stress. These experiments included CK3h (SRR3735179), CK6h (SRR3735193), CK18h (SRR3735547), CK24h (SRR3735572), SY3h (SRR3735589), SY6h (SRR3735674), SY18h (SRR3735739) and SY24h (SRR3735764) where “CK” refers to the group that served as the experiment’s control and “SY” refers to the group that served as the subject of the experiment. During various stages of drought 35 of the 54 possible TCS genes were found to be expressed while the remaining genes showed no expression. Only three genes, *VrHP6.1*, *VrRR3*, and *VrRR6*, showed higher levels of expression in the first three hours of drought exposure. However, as the duration of exposure increased, the expression of these genes decreased significantly. After being exposed to a drought environment for six hours, the genes *VrHK4.2*, *VrHP4.2*, *VrHP1.3*, *VrHP2*, *VrRR5.1*, and *VrRR5.2* showed upregulated expression significantly. *VrRR17* showed a slightly positive expression, while the other genes showed negative expression. After eighteen hours of exposure, the genes *VrHK1.1*, *VrCK12*, *VrHP1.2*, *VrPRR7.1*, and *VrPRR7.2* were highly upregulated, while *VrHK1.2*, *VrRR10*, *VrRR11.1*, *VrRR11.2*, *VrRR14*, *VrHP4.1*, and *VrPHYB* genes were moderately upregulated. The majority of TCS genes were highly upregulated after 24 h of drought stress exposure. Cytokinin-related genes *VrHK1.1*, *VrHK1.2*, *VrHK2*, *VrHK3*, *VrCKL2*, *VrPHYB*, and *VrPHYE* were found to be more expressed after 24 h of exposure compared to other time frames. Ethylene-related genes *VrETR1*, *VrERS2*, *VrEIN4.1*, and *VrEIN4.2* were shown to be upregulated. Only one histidine phosphotransferase, *VrHP4.1*, was upregulated. Interestingly, the genes that were upregulated after six hours were downregulated after 24 h. Positive expression was also found in genes related to response

regulators *VrRR10*, *VrRR11.1*, *VrRR11.2*, *VrRR14*, and *VrRR36*. These expression results show that as the duration of the stress increases, so do the expression levels of these genes, indicating that these genes promote plant growth and development under stress conditions (Figure 6A).

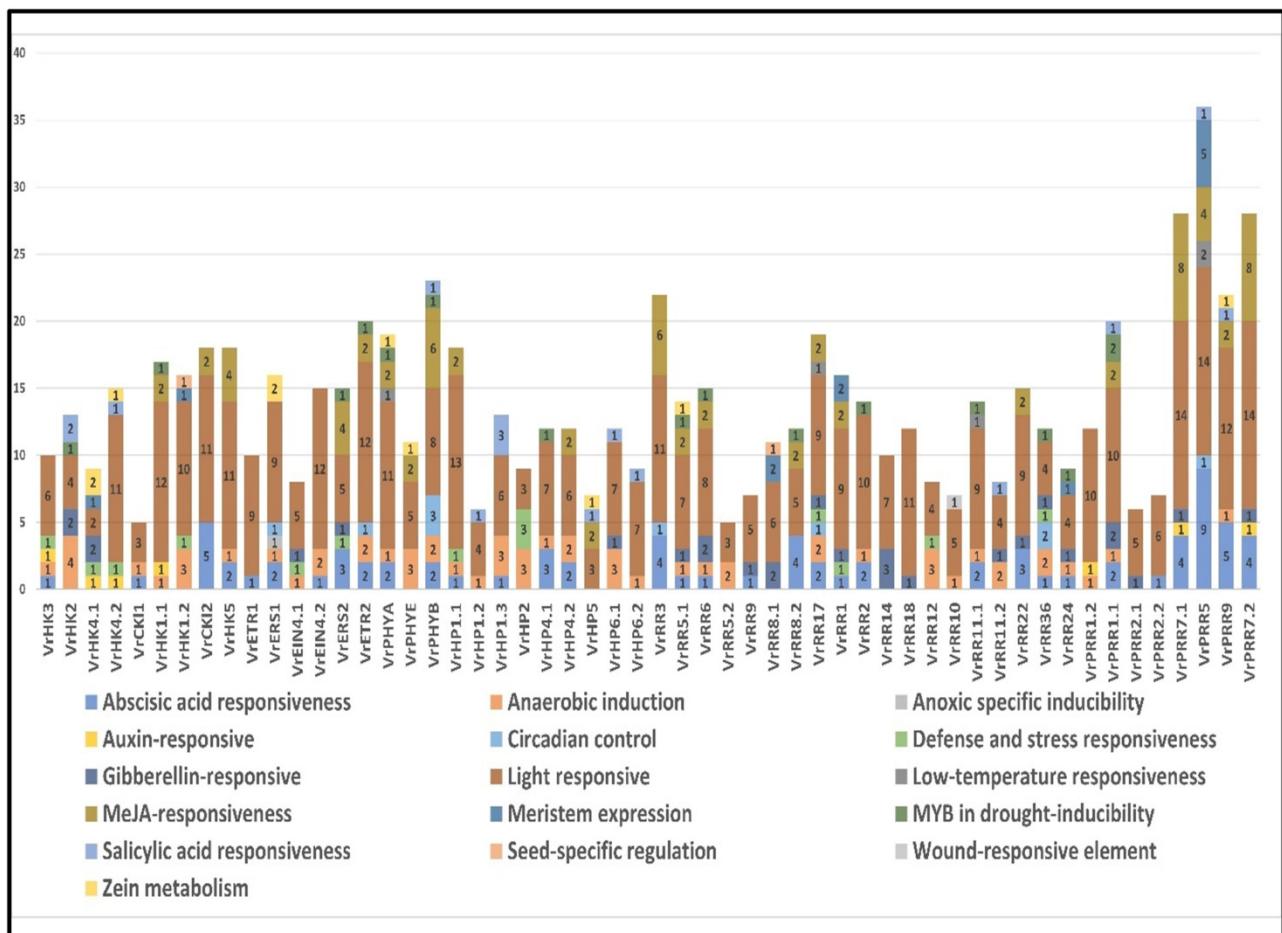
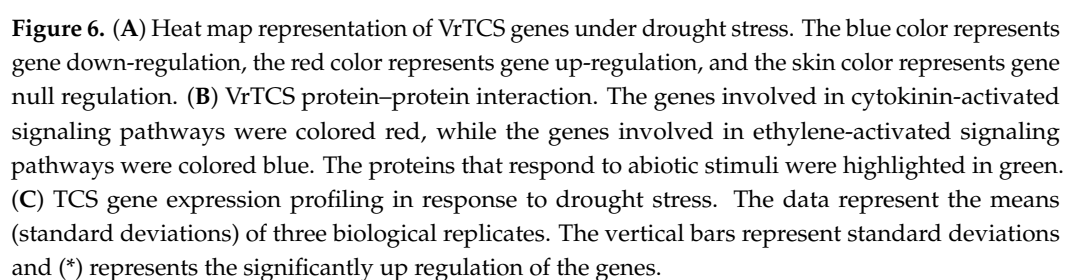


Figure 5. The cis-regulatory components of *V. radiata* are graphically shown. Various colors represent various types of cis-elements and the number on the bars shows the number of that element present in that gene.

To provide information on the expression profiles of two-component system genes in leaves in response to drought stress, we used real-time RT-qPCR in *V. radiata* leaves to examine the expression of selected genes. Based on the VrTCS RNA-seq data from 54 genes, 12 genes were selected that were differentially expressed. After three weeks of germination, all drought stress (0 days, 3 days, and 5 days) was applied to the plants. Drought stress altered the expression of the following genes: *VrHK2*, *VrHK3*, *VrHK4.2*, *VrPHYE*, *VrEIN4.1*, *VrHP2*, *VrHP4.1*, *VrHP6.1*, *VrRR3*, *VrRR5.2*, *VrRR10*, and *VrPRR9*. These findings revealed that the overall expression trend of these genes obtained through qRT-PCR analysis was highly consistent with the RNA-seq data except for *VrHK4.2* and *VrPRR9*. At 3 days of drought treatment, significant expression was observed in *VrHK4.2*, *VrHP2*, *VrRR5.2*, and *VrRR10* whereas expression of these genes significantly dropped at the fifth day of drought treatments. It suggests their role in early response to drought stress. Except for *VrHK4.2* and *VrPRR9*, the overall expression trend of these genes obtained through qRT-PCR analysis was highly consistent with the RNA-seq data. On the third and fifth days, the expression of the *VrRR3* and *VrHP6.1* genes was significantly reduced. The expression profiling revealed that as the drought duration increased, so did the expression of these genes (*VrHK2*, *VrHK3*, *VrHK4.2*, *VrPHYE*, *VrHP4.1*, and *VrRR5.2*), facilitating plant growth (Figure 6C).



The TCS interaction network identified three distinct TCS clusters as HKs, HPTs, and RRs. During the interaction, all of the HK proteins were linked to the HPTs proteins, which were then linked to the RRs protein. As a result, this TCS interaction network provides strong evidence that HPTs act as bridges or hub proteins between HK and RR in plants, transferring the HK phosphoryl group to the RR protein. TCS proteins have been shown to interact with one another, which has resulted in the discovery of several metabolic and regulatory pathways. Furthermore, all of the HKs, HPTs, and RRs in the interaction network provide evidence that they are involved in the cytokinin-activated signaling pathway, while ethylene-related proteins were involved in the ethylene-activated signaling pathway. Furthermore, some response regulators, such as *VrRR3*, *VrRR6*, *VrRR5.1*, and *VrRR5.2*, as well as histidine kinases, such as *HK1.1*, *HK1.2*, *HK2*, *HK3*, and *HK4.2*, were found to be involved in the response to abiotic stimuli that activated the signaling cascade (Figure 6B).

4. Discussion

Several environmental conditions can inhibit plant growth, development, and productivity. Sessile plants have evolved different signaling pathways to help them survive under different environmental conditions [9]. The TCS gene family plays an important role in signal transduction and, as a result, this system aids the plant's development and growth [1,3]. As a reason, identifying and functionally validating TCS involved in metabolic pathways and signal transduction may contribute to the development of such kinds of crops with enhanced characteristics, such as tolerance to stress, to address the challenges of global warming and climate change issues. These studies were conducted on a variety of model and non-model plant species.

TCS genes have been discovered in a variety of plant species, including *A. thaliana* [15], *S. bicolor* [59], melon cucumber [61], *C. arietinum* [58], *B. rapa* [68], *C. lanatus* [61], *O. sativa* [60], *S. lycopersicum* [64], and *G. max* [10]. In this study, 54 TCS-related genes were discovered in the entire genome of *V. radiata*. This number of genes is greater than the number of genes found in the genomes of *A. thaliana*, *S. bicolor*, *P. trichocarpa*, *O. sativa*, and *L. japonicus*, with 47, 37, 49, 37, and 40 genes, respectively, while the number is lower than the TCS genes of *S. lycopersicum*, *T. aestivum*, *Z. latifolia*, and *G. max*, with 65, 62, 69, and 98 genes, respectively [10,59,60,63,67,76]. VrTCSs are segmentally distributed on chromosomes. Only three pairs of duplicated genes were found. Numerous plants, including *A. thaliana*, *S. bicolor*, melon cucumber, *C. arietinum*, Chinese cabbage, watermelon, *O. sativa*, tomato, and *G. max*, have both tandem and segmental duplication in their genomes, suggesting that genome duplication is significant in gene family duplication. The Ks for segmental duplication in this investigation ranged from 1.585 to 2.545, corresponding to divergence times of 120.80 to 193.99 Mya. In tomato, the segmental duplication (Ks) value ranged from 0.79 to 0.60, with the tandem duplication period lasting from 5.96 to 26.55 million years ago and the divergence period lasting from 46 to 60 million years ago (Mya) [64]. The first duplication event in *C. arietinum* occurred 256.7 Mya ago, and the most recent random duplication occurred around 38.90 Mya ago, resulting in the emergence of a novel gene [58].

In phylogenetic analysis, VrTCS genes were found to be grouped into the same subgroups as those found in earlier investigations of *O. sativa*, *Z. mays*, *S. lycopersicum*, *G. max*, and *A. thaliana*. TCS subfamilies are present in all of these plants (HK, HP, and RR). As in *A. thaliana*, the conserved functional domains were employed to categorize the subfamilies in *V. radiata*. The Rec/Response domains in *A. thaliana* were used to distinguish between the types of RR family members. In *V. radiata*, the RR family clades shared the same domains [17].

The identification of cis-elements in the promoter regions of VrTCS genes assists in the discovery of switches that regulate gene transcription. Our analysis indicated that many drought-responsiveness, light-responsiveness, ABA-responsiveness, and hormone-responsiveness components were found to be associated with wound and stress responses.

Similar components related to the TCS were found in the promoter regions of previous plants studied. Light-sensitive and hormone-associated components (TCA elements, gibberellin, and ethylene responsiveness element (GARE and ERE motifs)) were widespread in the HK family of dicots plants. Both biotic and abiotic stress-responsive components are found in the RR family. A substantial quantity of comparable stress-responsive elements have been discovered in cucumber and watermelon [61], comprising ABA-responsive, ABRE, drought-responsive, MBS, and components. In addition to these stress-responsiveness elements seen in Chinese cabbage, type-A RRs also contain GARP binding sites [68]. The type-B ARRs may bind to these promoters, activating transcription. It has been observed that transcriptional regulation type-B RRs are partially required for the activation of cytokinin-dependent type-A RRs. This implies that in addition to hormone signals, other genes that respond to ripening or stress conditions may potentially trigger the TCS genes. In our findings, we discovered a large number of these associated switches in their promoter regions.

Drought, salinity, and temperature are examples of abiotic factors that may affect plant growth and maturation. TCS are involved in abiotic stress responses, and thus their relative expressions have been studied to further understand their function in adapting to environmental changes. *A. thaliana* contains the *AHK1* gene that helps the plant respond well to drought and salinity stress. In addition, the genes *HK5* and *EIN2* were responsive under saline conditions. In this investigation, the VrTCS gene family was shown to have a time-bound expression of drought stress. *VrHK2*, *VrHK3*, *VrHK1.2*, and ethylene receptors were found to be upregulated in seeds. *CarHK2*, 3, and 4 were found in all tissues of *C. arifolium*, whereas *CarHK1* and 5 were found in pods and shoots, and the response regulators were found in floral buds [58]. Members of the rice histidine kinase family were discovered to be expressed in the shoots/roots, while members of the histidine phosphotransfer family were discovered to be expressed in the leaves, and RR family members were discovered to be expressed in the stems, roots, leaves, and spikelets. Response regulator is expressed more in the melon's root, implying that these genes are involved in cytokinin signaling. Similar findings were also observed in *B. rapa* [68].

Agriculture is most vulnerable to two abiotic stresses: salinity stress and water depletion. The vast majority of available evidence suggests that TCS genes are involved in a variety of environmental stresses. This study identified 54 VrTCS genes, some of which showed a negative response to drought stress, but the majority showed a positive response. Drought stress reduced the expression of these genes in *A. thaliana* but increased the expression of HPs and RRs in *S. lycopersicum* and *S. bicolor*. *VrHK3*, *VrHK2*, *VrETR2*, *VrERS2*, *HP4.1*, *VrRR1*, *VrRR12*, *VrRR36*, *VrPRR2.2*, *VrPRR5*, and *VrPRR9* were upregulated in *V. radiata* under drought stress, whereas *A. thaliana* *ARR1* and *ARR12* were downregulated [18]. Some *V. radiata* genes related to response regulators (*VrRR3*, *VrRR5.1*, *VrRR5.2*, *VrRR6*, *VrRR8.1*, *VrRR8.2*, and *VrRR17*) were downregulated, and similar expression was also observed in *Cicer* response regulator 5 and 12 in drought stress conditions [58]. Only a few numbers of VrTCS genes were found to be silent. Expression analysis of drought environment depicted that the majority of TCS genes exhibit expression as the time of the water depletion increases. These findings of the TCS gene expression show how these genes respond to abiotic stresses.

5. Conclusions

In this investigation, we discovered 54 potential TCS protein family members, comprising 18 HK (L), 9 HPs, and 27 RRs. The categorization of proteins, domain architecture, gene structure, and evolutionary linkages with other species, gene duplication, and gene location on chromosomes were all carefully investigated. The sequence and domains of these TCSs were found to be highly conserved. TCS proteins from other plants seemed to have a closer phylogenetic connection with VrTCS proteins. Expression analysis of VrTCS genes revealed that some of the genes were expressed in early response to drought stress, while the majority of them were expressed in late response to drought stress. These

findings give us valuable information on *V. radiata* TCS genes that will assist in the functional characterization of signal transduction pathways and in improving the plant's stress tolerance.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/agronomy13040989/s1>, Figure S1: Protein sequence alignment of histidine kinase and ethylene receptors. The boxes represent the conserved phosphorylation residue. Figure S2: Protein sequence alignment of phytochrome family and histidine phosphotransferase proteins. The boxes represent the conserved phosphorylation residue. Figure S3: Sequence alignment of all types of response regulators. Showing the conserved residue in their receiver domain. Figure S4: Conserved motifs present in the protein sequence of VrTCS. Figure S5: Phylogenetic tree of histidine kinase protein with different species and different symbols representing the different species HKs. HK1 family, HK2 family, HK5 family, Ethylene receptor, and phytochromes were represented by blue, red, brown, purple, and green colors. Figure S6: Phylogenetic tree of Histidine phosphotransferase proteins with other species. AHP1-like proteins were grouped in a separate clade and represented by red color, AHP2 protein represent by green color whereas blue and purple colored clade show AHP6-like and AHP4-like proteins respectively. Figure S7: Phylogenetic tree of Response regulator proteins with other species. Type-A RRs-like proteins were grouped in a separate clade and represented by red color, PRRRRs-like proteins represent by green color whereas blue and purple colored clades show Type-B-like and Type-C-like proteins respectively. Figure S8: Representation of all the cis-regulatory identified in VrTCS promoter sequences. Various colors show a different types of switches discovered in each corresponding sequence. Table S1: List of primers used in this study. Table S2: Conserved domain and cellular localization of VrTCS proteins. Table S3: Cis regulatory elements discovered in promoter region of VrTCS.

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